

#033 Karl Friston - The Free Energy Principle

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[Music] welcome back to the machine learning street talk youtube channel and podcast so today we have an incredibly special guest professor carl fristan it was one of the most fascinating conversations we've ever had on street talk this is an old school professor he went to cambridge in 1980 and he's one of these kind of um eccentric polymath types that you know sits on the the old kind of chesterfield chair with all of the springs coming out and you know smoking a pipe or something like that professor fristen is most well known certainly in the machine learning domain for his free energy principle or active inference which is a kind of reinforcement learning flavor of that if you like but he's got an incredible background he is an expert in psychiatry and cognitive neuroscience and physics and bayesian statistics it's so strange just to have all of this knowledge embodied in one person so it really was quite an interesting conversation actually now the free energy principle has been called a postulate a natural law an imperative an unfalsifiable principle it's been called many things it aims to give an almost universal understanding between the mind life and the environment so how did the free energy principle come about the free energy principle as you have just described it started really when i was a student aspiring to put together maths and psychologies and gets back to mathematical formalisms the you know the principles that underlie the sentient behavior with which we are gifted and the product of that was the free energy principle the free energy principle sets the foundation for planning as inference by explicitly modeling the world and its states as beliefs it balances accuracy with entropy which maintains the flexibility needed to continually adapt to future outcomes and explorations but the more interesting game is i think better cast in terms of planning as inference enabling you to roll out much further into the future and ask well what would happen if i did that what would my beliefs be about the state of the world in the long term future so i think prediction in its full and glorious anticipatory sense really takes center stage features of reality itself such as self-organized behavior and even quantum they seem to require some kind of probabilistic bayesian belief update on world states for example the path integration formalism from richard feynman it essentially averages

over many probabilistically weighted paths in other words functions over beliefs and has proven crucial to the subsequent development of quantum electrodynamics that's what happened to understand the first principles underlie sentient behavior you have to understand the dynamics of self-organization in particular self-organization of systems that are open to the environment that comes in through Feynman's path integral formulation and thinking not just about the flow of the dynamics of self-organization at this point in time but trajectories into the future and the probability distributions over those trajectories and particularly the states that act upon the outside and then things get much more interesting you can interpret this in terms of inferring the most likely paths basically as resting upon a prediction of states of the world in the future conditioned upon a particular sequence of actions or policy the heart of the free energy principle and what sets it apart from alternatives is the strict balance between accuracy and simplicity evidence and entropy one of the things that I think is very interesting about the free energy formulation is that prediction is half the story so getting accurate predictions about the future while very important is juxtaposed with keeping your options open keeping a flexible mind keeping a high entropy model of the world so that as you encounter perhaps new situations it has the flexibility to adapt in the central role of relative entropies in this sort of variational construct I think that your former that is so important in minimizing the free energy you're also trying to maximize the entropy it seems sometimes counterintuitive but it is exactly that which is really mandated by things like Occam's principle and very practically relevant so if you don't do that if you don't put that uncertainty into the games then you're going to run into things like sharp minima and you're going to be searching for resolutions of that in terms of broadening your uncertainty flattening that free energy landscape just to try and secure those flat minima where you can be more reasonably assured that you've got some global minima say in standard deep learning or a machine learning context so this led Friston to believe that goal-directed behavior essentially planning based on goals is insufficient we need to have a more sophisticated system that can reason about the uncertainty of our beliefs we see echoes of this in machine learning we spoke to Kenneth Stanley recently about novelty search and in his formalism he explicitly avoids objectives and he thinks that we should use novelty because of the inherent deception in objective search similarly in classic machine learning we use regularization to stop the machine learning algorithm overfitting the training set Friston argues that the free energy principle combines all of these different paradigms we're getting quite close to the center of the bullseye here and we're talking about the dichotomy between belief-free and belief-based methods you know

you said in your papers that goal-directed behaviors is fine for learning kind of basic habitual policies of course normally there's this value function that needs to be either computed directly in the non-stochastic setting using the bellman equations or or some other method but but your approach is a stark departure away from having this value function again you've said everything that i could possibly say so i'll send you on my next lecture tour that'll take the pressure off me yeah absolutely and of course i forgot to just to highlight this sort of exploitation exploration dilemma that has done 20th century thinking minimizing this mixture that just is the the expected free energy you you've got exploration and exploitation sold for free in the right order so you normally see this food in the fridge before you start preparing your meal you don't do it the other way around and but you said it you said that the bellman authority principle is only fit for purpose if there exists a value function of states that will ensue if i connect to that action let's talk about markov blankets markov blankets are probably the one piece of jargon which you're going to hear today more than any other term we have an innate common sense notion of things and the boundaries that separate them however capturing this mathematically is more difficult than we might expect the markov blanket concept formalizes these notions and underpins the free energy principle one of the pivotal concepts in the free energy principle is markov blankets tell us about them they're relatively straightforward if you imagine um here on our screen we're showing an image and if every single one of these little disks that you see there was a state and if you have a system that can be partitioned then the red dots that are in the middle are the thing itself the blue dots that are outside are the universe at large the external environment and the yellow and orange dots in between form the marcotte blanket and what the markov blanket requirements are is that the red is able to interact with orange okay orange is able to interact with yellow and yellow is able to interact with the blue but what we don't have is we don't have blue being directly affected and vice versa by by red so we don't have this so instead we have this these boundary sets we partitioned all the state space into an internal group the red ones an external group the blue in these blanket states that sit in between the two and facilitate the interaction between external and internal and why these are important for the free energy principle is they set up these conditional independencies which say that the blue states as long as i know yellow and orange they're independent informationally from the red states and likewise the red states so long as i know the blanket states are independent of the external ones and yet future conceptions might rely on even more general concepts that allow for markov blankets like for example wandering sets i'd like to explore that as you put it maliciously engineered false

dichotomy a little bit further but in a different context and someone of the marcotte blanket for example you were with uh sean carroll uh long ago and i think he asked does a hurricane have a markov blanket and he said well no it doesn't and that's that's quite annoying actually if it's enough of a boundary enough of a blanket if you will i can still think about in useful ways is that true i mean do we really need an exact markov blanket or is it flexible enough that it can be quite a fuzzy boundary there's an excellent question and the honest answer is i i don't really know at this stage but what is known uh and in not knowing that i mean that in a positive way that this is a future challenge i think people will contend with in the next few years so in an idealized formulation the markov blanket exists at non-equilibrium steady state so it exists in eternity in a very crisp and well-defined way that there are specific conditional independences that allow for a vicarious coupling between the inside and the outside so it's the device that gets you out of 20th century equilibrium physics that we're not talking about closed systems anymore we're talking about uh non-equilibrium steady states so we're really in the sort of the things that people like sean carroll a lot of other people are contending with at the moment the physics of open systems and self-organizing systems and a markov plan kit plays an incredible incredibly crucial role here in demarcating the edge of the system as it were that allows you to identify something stuff on the inside the markov blanket has to have a kind of synchrony with stuff on the outside and in virtue of those conditional dependencies that stipulatively define the markov blanket you can now treat the internal states as parameterizing probability distributions or belief distributions about external states so there's quite a fundamental bit of information geometry that you bring to the table when you have a markov blanket that suddenly you can interpret the machinations on the inside in your computer or your variation auto encoder or your brain as having beliefs of a bayesian sword about what caused these data what's going on the outside so the markov blanket is absolutely central everything inherits from the markov blanket so to move on to your more challenging question does a hurricane or the eternal flame have a mark of black and strictly speaking no it doesn't because you've got this fuzzy leaking it is from the point of view somebody's wanted to simplify things as a physicist very irritating but of course it's all also incredibly important and challenging so where would you go to try and understand these fluctuating blankets where you have exchanges and i worry about this every time i cut my fingernails or have my hair cut at what point did my mark or blanket become an external state so there's clearly there's clearly a bit of work to be done mathematically to accommodate markov blankets as themselves dynamically random objects my guess is that you're going to probably go back to the work of irkov who

was this one of the key intellectual architects of ergot history in ergolic theory and he was at one point occupied by the notion of wandering sets so if we think of the markov blanket as a partition of all the states of the universe into lots and lots of particles that comprise the internal states and their mark of blankets and if you consider that partition into particles as a partitioning into subsets and then you bring wandering sets into play you can now start to see a mechanics and a mathematics where you can now actually explain things like hurricanes and candles 3d printers printing themselves is a species of 3d printers one big mark of blanket or do i drill down and just say markup that only really exists for the lifetime of the 3d printer so all of these issues i think speak to exactly what you're drilling down on which is how can you accommodate fluctuations and if you like physics of non-equilibria not of states of a universe but of the blanket or the partitions that define the other markov blankets does the free energy principle conform to itself if we can imagine into the future maybe and if in the spirit of the free energy principle we maintain uncertainty about the free energy yeah that's very clever yes yeah so i often say in moments of vanity and pride that the free energy principle is one of the very few principles that conforms to itself it's trying to provide an accurate but minimally complex explanation for everything and the only other thing that i know that conforms to itself is the principle of natural selection in the sense that the theories of natural selection themselves evolved from perry and third world and in that sense this notion that the free energy principle should accommodate a judicious amount of uncertainty about itself is absolutely right yeah that's a great very clever observation fristen would argue that certain states of the world and certain states of knowledge would kind of transform themselves conforming to probability theory the free energy principle implies that non-equilibrium steady-state systems must directly model such states of knowledge and maintain them according to the rules of probable inference in order to continue existing in other words survival fitness requires inference we had this very interesting point from conor about there may be game theoretic pressures let's say to drive down accuracy in in some sense the free energy principle is a lot about the idea of we want to gain information to create a better more accurate model of the world around us but there are situations in game theory and decision theory where that's actually not fitness enhancing you might not want to learn the face of your kidnapper because he's more likely to kill you yeah that's a fascinating area and i i see those those sort of themes in many different contexts i never heard that before about not wanting to know the face of your kidnapper but that's a beautiful example it really does tell you that objective functions have to be about beliefs and the consequences of belief states not states of the world so if

there's one example that tells you you're not going to serve up with the bellman optimality principle that's going to be it but it also tells you that many of these real world problems have to contend with the fact that the kind of external states that you are dealing with are composed of sentient creatures like and that they also have beliefs keith why don't we talk about the main formalism for the free energy sure so the free energy principle comes down to the equation that we're showing here on the screen which is that the free energy is the summation of of two important pieces the energy which is a measure of how well your model fits the observations or the evidence that you're observing and then the second piece which in a lot of ways is really the more important piece of the free energy is the entropic contribution and so that's the kl divergence if you will between your model of the world and then the actual uh hidden model of the world and the divergence between those two it has two important pieces to it and i want to kind of call out i want to highlight one here which let's circle right there that's purely the the model entropy okay so that's actually the the negative of the model entropy and the in the kl divergence and so as the model gains greater and greater entropy and note that that's a positive quantity there okay then up in the free energy it's being subtracted from the free energy so the higher the entropy of your model the lower the free energy so as you're trying to minimize free energy you've got two competing criteria one is to fit observations better and better but that will require more and more complex models which will have lower and lower entropy and therefore they will be subtracting less from the free energy so what you're trying to do is find the that fits the data well while also maintaining a high entropy and the reason why that's for survival is that if you have a higher entropy model it's maintaining greater flexibility to to incoming information so if you recall log evidence and any associated bands like free energy or an elbow edwin slow bound in machine learning can be written as a as the accuracy minus the complexity and the complexity is just the divergence between your posterior and your prime belief distributions so that is a really and possibly the more important part of the free energy the accuracy is well understood but doing it in a minimally complex and maximally compressed way that that's the heart of it so recall the complexity is just the relative entropy between the prior and the first year it is the degree to which you change your mind in the face of this new data or this new sensory evidence the free energy principle equally weights accuracy and complexity might it be the case that we need to have a temperature parameter like a kind of knob to tweak the relative contribution accuracy and complexity would such a parameter be useful i'm wondering is in the free energy principle term with the complexity i'm wondering if maybe there should be a multiplier there like an

alpha something like a boltzmann constant that allows me to tweak the relative balance between accuracy and complexity there are two answers to that question first of all absolutely not the whole the whole point of dissolving that exploration exploitation dilemma the whole point of putting the information gain in the same space and in the same currency and on the same footing as your log prior preferences your reward your utility your belmonesque like imperatives is that there is a seamless exchange in terms of gnats natural units between the decomposition of your expected free energy in terms of this intrinsic value and this extrinsic utilitarian pragmatic value the other answer is absolutely yes but in order to in order to acknowledge that if you're just trying to explain the necessary properties dynamics of systems that self-organize to some non-equilibrium steady state you are saying nothing about the nature of that city state other than it is at steady state so it could be very high entropy steady state it could be very low entropy steady state it could be very hot it could be very cold you haven't really committed to any kind of steady state which means that it's slightly disingenuous to say that the imperative for everything is to minimize say a free energy functional and if you parameterize up with a particular parameter and we'll call it alpha as a not to your question then suddenly you really do need this alpha and this alpha gets in exactly as you say as a knob on the expected complexity versus the expected accuracy or ambiguity that's not in the literature so there are ongoing debates amongst the younger people who love the maths of this about whether we just need a generic k or whether we need to exclude bits to get to expected free energy you have to really think about the relative importance for this steady state of technically the risk and the ambiguity and the sensory entropy so that's again that's a bit of an open question which i'm hoping will be resolving about a year's time at which point your alpha will occur and i'll try to call it alpha in your honor one of the previous guests to our show dr hari valpola the ceo of curious ai and a computational applied neuroscientist he centers in a couple of questions for fristen actually and the answer was quite interesting but the crux of it is the brain is highly specialized right is the free energy principle an oversimplification for what's going on in the brain if you take a young child around nine months old the critical period for learning the phonemes of your mother tongue and you play some speech from the radio the result is nothing but if you play the same speech during a social interaction the child will learn to discriminate the phonemes in the speech structure matters structures of the being very finely attuned to and adapted to the context in which they're that they're making their inferences what would normally cast that in terms structure learning also known as bayesian model selection so this notion speaks to a simple

understanding of evolution or as nature's way of doing natural bayesian model selection i.e natural selection where you operationally associate the adaptive fitness with the probability this phenotype exists which is just the evidence that it is there the probability of finding that phenotyping place you know but it is on some reading provably true that for any system or agent to regulate its environment it has to embody or be a good model of that environment what does that mean its structure must somehow recapitulate the structure of the environment generating that it has to control or it has to engage with so it is hardly unsurprising that the delicate deep hierarchically structured connectivity we find in their brain and in a variation auto encoder is a natural thing that has emerged from the evolution of these architectures that all are conforming with the principle of free energy minimization what about daniel kahneman's system one and system two of cognition do those fit into the free energy principle he cites kahneman system one and system two of thinking there's this famous test where if you give monkeys a sequence of cards with a hidden pattern on which need to be classified into two classes humans suddenly click they see the hidden rule because their system two kicks in and they start getting 100 accuracy very quickly but you don't see that with monkeys so valpolia was saying that he doesn't see how that phenomenon could be explained by the free energy principle recent thinking about the cerebellum is that it plays the role of actually a supervisor so it may well be the case that the cerebellum plays a role in the amortization of carefully acquired skilled movements that become increasingly skilled as the cerebellum watches the cortex in that sense the cerebellum can i think very usefully understood in terms of being involved in supervised learning but i'd actually turn it on its head and basically say it's been supervised by the cortex but in a way that allows cerebellum to tell the cortex well normally you do it like that do we do our belief updating and a bit of planning as inference or do we just do what we've always done habitually respond quickly and efficiently by harnessing something that has already been amortized and that i think is a really interesting interpretation of habitization versus deliberative thinking which from the point of view of a neuroscientist would be the equivalent of system one versus system two in that sort of edge between machine and reinforcement learning sometimes referred to as model-free versus model-based so it's the bread and butter of a jobbing neuroscientist certainly a systems neuroscientist to the computational architectures and the nature of message passing implied by either belief updating or predictive coding or variational message passing how it is enacted physiologically on a neuroanatomy that has a deep structure connor asked fristen about predictive coding there has been a paper that says that predictive coding can approximate back

propagation in arbitrary graphs i'm sure a friend of mine has contributed that and if there's any sense that prop has been demeaned as something which is not that's really central to everything then i think that's probably wrong so from my perspective it is in very the case that the gradients with respect to anything of variation free energy can be written down as an a prediction error which means that if you believe that the universe of interesting things is performing a gradient flow on a variational free energy functional then you also believe that they are by the chain rule minimizing their prediction error i think that's one of the first times that premier scientists actually say something positive about backdrop and does the free energy principle tell us anything about the origin of life for example abiogenesis if we take for granted that the free energy principle explains a great deal about how certain systems of the kind we've been elucidating will function does it explain the origin of those systems is it almost unavoidable that we will end up with these deeply delicate markov blanket systems that become increasingly rich and have longer longer information lengths and more itinerant dynamics as the universe unfolds it would be great to know the answer to that and to have a tractable mathematical formulation of that i'm afraid that the free energy principle in its vanilla form as it currently stands does not give you that no all it says is if that thing has evolved these are the properties that it must possess but at least it can tell us we should continue exploring and looking around to address that uncertainty yes cover yes of course some better better free energy principle next we're going to talk about richard feynman's path integral this is something that fristen referred to many times during the session today yeah so path the path integral formulation let's say of which is essential for quantum electrodynamics for one thing is quite interesting and actually it should be familiar in in one sense to to let's say bayesians or people who do a lot of conditional probability theory because they may recall that the probability of say a particular event is actually equal to the summation of the probability of that of that event given like let's say some other state of the world okay over all possible states of that a in other words this is marginalization or the marginal probability if you have a bunch of possible worlds what you have to do is sum over all possible worlds and when that's applied to physics say in the case of quantum electrodynamics what it's saying is that if you have many possible ways in which let's pretend the photon could come from a and end up over b it could travel along any one of these many paths here that if you're considering the total probability of that actually happening you have to sum over all these possible states weighting by their amplitude and that's a very beautiful consequence because in some real essence we don't think that the photon say travels every one of those possible but it's as if it did and why this is

important in the free energy principle is that when we live in a universe where you have to perform this summation over all possible worlds that means that inherently we need a system that can model probabilities and can transform those probabilities accurately according to the laws of probability theory and another interesting consequence of it is that if we're playing this repeated game where we're trying to survive over time and we're getting in new information all the time about these many possible future paths we have to model the world in a flexible way we have to keep an entropic model a model that has a high entropy that's able to adjust as new information comes in about those probable paths how did he find it talking to uh professor fristen oh i i really i have to say i very much enjoyed talking to professor fristen when i went into the preparation for this call some weeks back i would say i was probably more skeptical of the free energy principle than i am now but as we've learned a lot over the last few weeks and then really having a chance to pose some questions uh to professor and hear his answers uh yeah i think this is a very interesting principle uh and i think it's i think we'll find increasingly as we become more sophisticated with both the modeling of the principle and the application of the principle that we are going to find many systems that that conform to it so i find it quite interesting and completely enjoyed talking to him i wish i lived next door to him i'd probably hop over there once in a while to have a sherry and just discuss deep topics he has such an intuitive understanding of deep topics that it's sometimes hard to keep pace but if you can it's it's very fun yeah because if you look at the free energy principle on wikipedia it even says at the top of the article that it's very complicated even experts in the field just have trouble getting their heads wrapped around it and when you listen to professor fristen talk he clearly is a polymath he has deep knowledge of so many different fields and it's really interesting to have all of that embodied in one person because the level of cross-pollination is really quite impressive but by the same token there's so much jargon and he's diving into so many different areas that it does make it quite difficult to follow what he's talking about i think that wikipedia has got it wrong that it's complicated i think that we often make the mistake of thinking that things that are hard to understand are inherently complicated and i think that's wrong i think simple and easy are very very different concepts it's possible for simple concepts to be very difficult to understand and indeed to take human civilization thousands of years to discover and i think this i think the free energy principle falls under the category of simple but very difficult to understand very deep deep concepts almost like some of the concepts that rest at the foundation of mathematical logic or probability theory you know when you start to think about puzzles like this sentence is false right or if you think

about what is the meaning of probability is it a frequency is it a degree of rational belief why or why not you know these are very deep concepts at the end they can be expressed relatively simply on paper but still be very difficult to thank you very much they're very informed questions i got a sense you knew the answers before you asked them that's how i felt anyway we really hope you've enjoyed the episode today we've had so much fun making it remember to like comment and subscribe we love reading your comments and we'll see you back next week the like new yorkers are rude and blunt you know they're they're they're very direct and they'll be about it well germans are polite but blunt like germans will very politely tell you your code is completely unacceptable but they'll be very polite about it that's kind of slightly similar to uh southern culture or classic southern culture in the us it's very polite but straightforward like they don't like beating around the bush you know but i would say they don't go as far to the rude spectrum while being blunt like they still are straight forward but politely i've i'm super sensitive to people beating around the bush yes that's it that's why i was funny that when you told me you were like a traitor in new york i was like yeah this makes a lot of sense like straightforward and polite i love germans i worked with a few of them in my lab and in graduate school and i like i also like how rule-based they are like there are rules and you follow the rules when you open the door you close the door you close it you don't leave it half open it's like i love it that's true it's true it's both a blessing and a curse i used to work with some radioactive dyes and i disposed of one in the wrong container so i disposed of like the betas in the alpha container or something and it caused like a cost like a couple hundred dollars or something the german postdoc there was like pissed at me but polite he's like just keeps going on and on about how can you put a in the b container right that's just not how it happens that should have never happened i don't know what we should do about this but something needs to happen to make sure you never do that again yeah that is such a german thing i i've heard that exact conversation with people before he would keep just wouldn't let this go i'm like yeah i'm sorry i screwed up you know it wasn't a good enough explanation it was like what is broken that we need to fix so this never happens again that's the most german thing i've ever heard hello hello hi there hello hello professor fristen great to meet you nice to see you fantastic now i'm not an expert on etiquette is it dr fristin or professor fristen which is more correct in england it would be professor pristin okay in the states i think i've heard that teachers are called professors so they prefer to be called doctors right in the uk it's a professor gotcha and tech industry is first names only [Laughter] welcome back to the machine learning street talk youtube channel and podcast with my two compadres connor leahy and mit phd dr

keith duggar now today we have a very special guest he is a british neuroscientist at university college london and an authority on brain imaging he studied natural sciences physics and psychology at the university of cambridge in 1980 and completed his medical studies at king's college hospital in london a strong indicator of professor fristen's illustrious career is that he's been cited over 206 thousand times with an aged index of 239 in 2016 he was ranked the most influential neuroscientist on semantic scholar and carl fristan pioneered and developed the single most powerful technique for analyzing the results of brain imaging studies he's currently a welcome trust principal fellow and scientific director of the welkin trust center for neuroimaging his main contribution to theoretical neurobiology is the variational free energy principle also known as active inference in the bayesian brain the free energy principle is a formal statement that the existential imperative for any system which survives in the changing world can be cast as an inference problem the probability of existing as the evidence that you exist if you will you can always interpret anything which exists as being separate to the environment it exists in carl asserts that existence is a kind of interface between the interior and the exterior so the free energy principle which is closely related to the bayesian brain hypothesis is a simple postulate with complicated implications any adaptive change in the brain or indeed any system which exists will minimize surprise or free energy as cold puts it this minimization could be on any time scale or on any system which resists a tendency to disorder from single cell organisms to social networks the bayesian brain hypothesis states that the brain is confronted with ambiguous sensory evidence which it interprets by making inferences about the hidden states which caused the sensory data so is the brain an inference engine the key concept separating fristen's idea from traditional stochastic reinforcement learning methods and even bayesian reinforcement learning methods is moving away from goal directed optimization belief based updating combines the ambiguous information with prior beliefs about the nature of the world the missing information problem is something which dogs many areas of machine as we discussed on our gpt3 episode last week implementing the bayes rule directly is often computationally intractable and computer scientists drew inspiration from the physics world for creating approximate inference techniques called variational bayesian methods and carl's active inference method uses these techniques to great effect anyway professor fristen it's an absolute honor to have you on the show welcome and how did you come up with this exciting principle well first of all let me congratulate you on that beautiful introduction you said everything that i could possibly say in the next hour and a half so i'll just try and recapitulate what you said the road to the free

energy principle as you have to put together maths and psychology so in those days it would have been known as mathematical psychology nowadays computational neuroscience and leading of computational neuroscience into machine learning so from the start that was the ambition more practically though as your brief resume indicated i got a bit distracted by becoming a doctor and then a psychiatrist in you out of clinical compassion but also partly out of an interest in understanding how the brain works and then inevitably one gets back to mathematical formalisms the the the principles that underlie the sentient behavior with which we are gifted and the product of that was the the free energy principle or active inference in a sort of more cognitive neuroscience setting i'd like to ask you about two kind of related topics that have interested in machine learning lately is i was interested if you were familiar with the work of jeff hawkins from numenta who is a unsung hero among machine learning inspirations and he had a quote saying it is the ability to make predictions about the future that is the crux of intelligence and recently with uh machine learning progress such as with gpt3 which you know is a better and better predictive model and some would argue including myself also becomes more and more intelligent in a meaningful way so my question is prediction really all it takes for intelligence or is there more to it i think prediction figures very centrally in the sort of dynamics that we're talking about can answer that question from two perspectives often cast in terms of the high road and the low road to that formal understanding of things like active imprints so that the low road would really be a bottom-up approach thinking what does the brain do and on that view or on that approach the the big move in the 21st century has been towards predictive processing and it started with predictive coding as a nice metaphor for message passing in the brain now generalized to subsume action and decision-making and choices and epistemic foraging and so andy clark coined the the phrase predictive processing to it to accommodate that and in that sort of formulation of sentient behavior prediction is absolutely essential at two levels and from the point of view of predictive coding obviously it's baked into the title that you're you're trying to predict what you will see under some belief about the sort of latent states generating some data or you could formulate that in terms of compression minimizing message length and efficient making sense of data or unpacking data the prediction in that sense is i think not quite the kind of prediction that you're asking about which is uh which has this sort of temporal aspect sort of the forecasting or anticipatory aspect of prediction however once you move predictive coding into a sort of bayesian filtering setting and you put dynamics in play then when you're trying to predict the trajectories in the moment you are implicitly predicting a short-term future so i

think sort of prediction in the dynamical the temporal sense starts to bite again beyond that there is in the neurosciences an appreciation that that's not the end of the game you can get a long way on this low road of understanding in terms of understanding how we act and perceive and how action is in the service of perception vice versa through treating motor behavior responses of an embodied brain in terms of reflexes and this would be very much a explanation of the action perception cycle essentially equipping predictive coding with reflexes of this sort of in the moment sort yeah you could write down in terms of differential equations or kalman bc filter but the more interesting game planning as inference so actually including what i'm doing as a random variable in your inference problem and enabling you to roll out much further into the future and ask long term future and that gives a very different aspect to prediction that you're predicting the consequences of any move on the world of any data that you might want to sample how will that reduce my uncertainty what information will i gain what relevance does that have for the kinds of preferred outcomes that characterize me as a goal-directed creature so i think prediction in its full and sense really takes center stage and when people talk about active imprints that sort of implicitly they're talking about this sort of fuller deep temporal approach to inferring what should i do next and then from those inferences selecting the next move to to make so in that sense the other sort of the hierarchical temporal models that dalip and jeff hawkins have have been talking about for you know decades as well and i can remember reviewing as a handling editor the first publication plus computational biology on the on this work and it was refreshing and i think that that's exactly uh the kind of hangers inference in the future consequent on or conditioned upon the different kinds of policies or sequential uh sequences of actions that i could entertain bringing time for center stage in into the inference problem the high role i should just add so using a completely different language now that i would use as a physicist to promote the importance of prediction it takes a slightly different view it says well to understand the first principles that underlie to the environment to their eco niche to their heat bath their heat reservoir and when one does that you can use uh dynamics and random dynamical systems to say the kind of moves and and changes of the systemic states must have this form of predictive coding or bayesian or kalman like filtering in the moment and that's relatively straightforward to show and there was a hint of the requisite mathematical apparatus you needed to make those assertions in the introduction and specifically mark off blankets that separate me from not me so that would be grade one if you like uh prediction it's just that we have dynamics and the dynamics required a self-organization to non-equilibrium steady state require some in the moment kind of

prediction so where is the sort of the the higher end anticipatory of dynamics well that that comes in through fyman's path integral formulation and thinking not just about the flow of the dynamics of self-organization at this point in time but trajectories into the future and the probability distributions over those trajectories and particularly the if you like formally characterize what we do over extended periods of time so when you bring the path integral uh formalism to bear on these distributions then you can sort of say what kinds of policies as i move into the future are more likely and which are less likely and then things get much more interesting and we talk about sort of ambiguity and risk and what the imperatives are for long-term behaviors and underneath all that there is an interpretation that i'm really trying to predict or you can interpret this states of the world in the future conditioned upon a particular sequence of actions or policy one of the things that i think is very interesting about the prediction is half the story so getting accurate predictions about the future while very important is juxtaposed new situations it has the flexibility to adapt more quickly that incoming information and that because it's a repeated game you're not just trying to optimize the very next step only but the entire trajectory of your existence if you will and and that's captured in the free energy principle inside the kl divergence part of that is the entropy my model of the world and the larger that is the lower the free energy is this interplay and you also point out in a video too that when it's this repeated game it's related in a way to what in other fields is called the exploration versus exploitation kind of trade-off like in the multi-armed bandit world where i can take an action that's maximally exploited but but i may not learn very much and then on the other hand i can take an action that's purely just to learn about the world but may not achieve much of my other objective and what i love about that is it explains why we're always so tempted to push the button why people always want to press the button and see what happens or go out and explore even if there's no immediate benefit and it has so many connections to things like novelty search and multi-arm banded analysis it's quite interesting yeah you bring so many fascinating points there yeah and under robotics intrinsic motivation and robotic artificial curiosity and the schmidt hoover-like sensor which itself all comes back down to this minimizing the complexity and paradoxically and celebrating so if you allow me you've made so many good points there i i can't resist just highlighting the first of all predictions only half the story absolutely i mean why are we predicting we're predicting in order to infer what to do next so the doing the action the inactive aspect really becomes essential thing in terms of well things that are written into the description like self-organization how i organize myself how what i do and if you're in data science that would be

basically how do i optimally update a mine can i find asian optimum design principles written into the formalism the variational formalism that the free energy principle so i think just saying that perception is great if you're just studying the visual cortex or you're interpreting some given pixelated image but the real challenge is really how do you go and take that picture and how do you move your eyes around and sample that information that's going to make much more sense in the world so it's all in the service of what i'm going to do you mentioned that the kl the central role of relative entropies in this sort of variational construct i think that your for me that is so so a very simple perspective on free energy which i think is useful certainly for students is it just to remind them that the free energy is some expected potential some expected utility if you like and complemented by the entropy and in minimizing the free energy you're also trying to maximize the entropy we've seen sometimes counter-intuitive but it is exactly that the game so you don't commit to a very precise explanation for these data then you're going to run into things like sharp minima and you're going to be searching for broadening your uncertainty flattening that free energy landscape just to try and secure those those those flat minima where you can be more reasonably assure that you've got some global minima you know that would be a story you could tell just about the composition of why free energy rests upon entropies and relative entropies but you took us um straight to the heart the sort of deeper in time inference that comes along with the active inference which is this sort of this implicit imperative to reduce that can often be described in terms of responding to salience or epistemic affordance to say if i did that what would happen what would i know what information would i gain novelty you mentioned so technically what we tend to do is to talk about the resolution of uncertainty or the information gain that is formally described by kl divergence between my beliefs about states of the world with and without those sensory samples or observations that i would get if i did that so that kl divergence can be applied to unknown states the world and that would be driving behaviors like looking over there to make sure that my hypothesis that was a butterfly or a bird or a candle was the best explanation for these sort of peripheral visual sensations but you've also got unknowns for example in a deep learning scenario the the connection strengths you know in the brain parameters that underwrite our brain connectivity that encode the lawful contingencies and statistical regulations of likelihood mappings or probability transition matrices so these don't change quickly in time but this still are equipped with a posterior belief or some some stochastic representation about which you can reduce uncertainty and we call that novelty so if you can get yourself into a situation where you reduce your uncertainty about

the contingencies and the parameters of your generative model then you're responding to their again an epistemic affordance there is exactly this a resolution of uncertainty formally we would write that down as an expected free energy i don't know if this helps but one way i find useful to think about this and indeed explain to students is free energy is a bound on log evidence simply the probability of some data under a model of how i thought those data were generated and you can always split that into accuracy and complexity so if i have a statistician and i wanted to minimize my physics free energy or maximize my machine learning free energy which is because it's just an evidence lower bound and so you'd find in a variational auto encoder what am i doing if i decompose the log evidence into accuracy and complexity what i'm trying to do is minimize the complexity of my accurate expectations and then we get back to minimum description lengths the underlying say you know monograph like complexity formulations of your most efficient coding and the like so you've got this way of looking at self-evidencing in the sense of just forming good beliefs of how my data were generated in terms of providing the simplest but at most accurate explanations or accounts of the data at hand so if we come back to this to the previous discussion about what would happen in the deep future so i did a deep research over all the sequences of moves that i could make what would the expected accuracy and the expected complexity look like and if you write that down they come out as things like risk and ambiguity so the expected accuracy or more precisely expected inaccuracy the negative accuracy the negative probability of your data under beliefs about their causes becomes ambiguity becomes the conditional uncertainty if you like about the observations if you knew them and the the expected complexity becomes risk or it has exactly the same form that you'd find in kl control theory or risk sensitive control in economics it's just the relative entropy or the divergence between what you think will happen if you did that and what a priori you think is going to happen which normally people treat as a sort of the preferred outcome the kind of things that a creature or a system or an agent like me would normally encounter so that is actually the thing that you were talking about the reason why we push that button or open that door or go on that trip or go into that google page you know it written into it is if you like the homolog of our aspirations for accurate accounts of all the sensory and other data that we have to assimilate when we consider the expectation in the future conditioned on what would actually do we're getting quite close to the center of the bullseye here and we're talking about the dichotomy between belief-free and belief-based methods so you've been sketching out this really interesting idea that we can take subsequent actions by maximizing the expectation over a a generative model

of states in distribution and subsequent policies but the alternative is so-called goal-directed behavior now you've said in your papers that you cite this wonderful example with an owl you say that an owl doesn't necessarily know where the mouse is in the field there might be some ambiguity there might be some and using a simple goal directed policy the owl would just go straight to the mouse and try and eat the mouse now it's quite interesting to contrast your approach to other forms of let's say bayesian reinforcement learning or or some other method but but your approach is a stark departure away from having this value function absolutely and of course i forgot to just to highlight this sort of exploitation exploration dilemma that has dog 20th century thinking of course that just dissolves in the 21st century because you know risk and ambiguity sometimes you can exchange the the terms in this expected free energy formulation uh and re-express that as an intrinsic um and extrinsic value where the intrinsic value is something that people in robotics artificial curiosity would recognize as that's of epistemic information gain uh part and the extrinsic value is just the expected utility of the expected reward and in minimizing this mixture that just is the expected free energy you you've got exploration and exploitation start preparing your your meal you don't do it the other way around so that's you know what you were hinting at in terms of the owl opening its fridge and searching it's slightly disingenuous of me i think to use such a stark contrast dialectic between what is in essence a whole bunch of things predicated on the bellman optimality principle and contrasts that with another way of doing it which is the which is essentially a variational hamilton's principle station reaction so the free energy principle inherits from the physics of non-equilibrium that you will find from thymann's path integral formulation of quantum electrodynamics right through to hamilton's equations of your motion so there are these this is the variational principle that is the first principle that we were mentioning before so the question is do you choose between variational principles of station reaction or do you go for optimality principles and in that choice what's the big commitment you're making you said it you said that the bellman optimality principle is only fit for purpose if i commit to that action whereas the variational principles of least action being a path integral of an energy and an energy being a log probability of a probability distribution immediately tells you that the objective function if you want to cast it in terms of normative theories of optimization the objective function is a function of a probability distribution so it has to be belief based a problem if if read a probability distribution is a bayesian belief so if you allow me to talk about beliefs simply as uh conditional probability distributions in the sense of belief propagation or bayesian beliefs then we know that the the from physics the objective function

the lipoon off function the cost function has to be a functional function of bayesian beliefs so that comes back to your point that if you go down the bellman route the rl route the optimal control route then you are predicting everything on a value function of states if on the other hand you go down a variational or a physics route then your objective function has to be a functional of beliefs about states and that's where you get the resolution of uncertainty that's where searching gets into the mix because to search is to resolve uncertainty but and surprise and all of these other attributes of belief updating are all attributes of probabilistic beliefs probability distributions so you know you have to have a function of probability distributions in order to account for sentient behavior and i would say including goal directed behavior why is that disingenuous well it's disingenuous because of course what you can do is take this more general formalism this variational formalism and take out of it all the uncertainty and then you can get back to the bellman optionality principle so really there's not a dialectic here it's just that the optimality principle deals with a special but very ubiquitous and pragmatically very important case where you can discount various sources uncertainty so common examples here let's ignore the partially observed aspect of a problem let's assume that our observations tell us everything we need to know about latent states causing data or latent states of the plant that i'm trying to try to control so that takes ambiguity out of the before before you do anything what you're left with where you're left with the expected risk what's that's the unexpected complexity so that's just a probabilistic measure relative entropy measure of the distance probabilistically between your prior preferences your goals your your rewarding states and what you think will happen so if you say look now let's assume for simplicity that it doesn't matter every which way that i can do this the uncertainty about what's going to happen is not going to change that means that risk now doesn't play any role anymore other than to um score the expected um utility or the expected log of your prior preferences that just is bay's bayesian decision theory so you can get quite easily back to what was a deliberate but maliciously engineered dichotomy you can sort of dismiss that just by acknowledging that you only need to worry about uncertainty and epistemics when you put uncertainty and not knowing your into the mix i'd like to i'd like to explore that as you put it in a different context and it's one of the marcotte blanket for example you were with uh sean carroll not long ago and i think he asked does a hurricane have a markov blanket and and he said well no it doesn't and that's that's quite annoying actually that it doesn't have a conv or maybe the eternal flame is sort of the the nemesis it doesn't we get to this when is a pile of sand a pile of sand what i'm curious about is does the concept of a markov blanket for the purposes of uh contemplating the free

energy principle can it be a fuzzy marcotte blanket so you know just for an inanimate example i can imagine a markov blanket around the moon and one around the earth but on the other hand they gravitationally interact but i don't think that precludes me from thinking about dimensions of the free energy principle even if even if there are interactions that maybe i that pierce the boundary if you will but still useful ways is that true i mean do we really need an exact markov blanket or is it flexible enough that it can be quite a fuzzy boundary what what is known uh and in not knowing i mean that in positive way that in the next few years and i'll and i'll try to explain why that's important from my perspective the the crisp and clear consequences of having a markov blanket is when one unpacks it right through to the path integral formulation the implications for the good trajectories in terms of minimizing expected free energy really do rest upon that being a well-defined markov blanket which as you say is the mechanism that allows for these connections and influences at a distance that would keep the moon in orbit around the earth for example whilst at the same time accommodating the conditional independences that enable me to distinguish the moon from something else so in an idealized formulation the markov blanket exists at non-equilibrium steady state so it exists in eternity 20th century equilibrium physics so are contending with at the moment and the physics of open systems and and a markup plan plays a incredible and it allows you to identify something so anything has to have a markov blanket strictly speaking for a particular period of time just as an aside what happens when you do that is stuff on the inside the markov blanket has to have a kind of synchrony with stuff on the outside and in virtue of those conditional dependencies that stipulatively define the markov blanket you can now treat the internal states as parameterizing bit of information geometry that you've been to the table as having beliefs of a bayesian sort going on the outside so that's absolutely crucial for the whole sort of active inference and certainly representational or realist or anti-realist sort of philosophical interpretations so the markup blanket is absolutely central blanket so everything else was in play we have the vodka plank equation we have the scrolling wave equation we have variational principles of stationary action and pathological formulations that's all there all you need to do is to drop the markov blanket in and then you get active inference so to move on to your more challenging question have a mark of black blanket well strictly speaking no it doesn't have exchanges so just technically you've got external states and then you've got blanket states that intervene between the internal states and the external states and yet with something like a candle flame or a hurricane you seem to have this exchange that what was once an external state becomes in a blanket state and what was once a blanket state becomes

an external state and i worry about this every time i cut my fingernails irkov who was one of the key of ergot history and ergodic theory and start to see a mechanics and a mathematics where you do actually can now actually explain things like hurricanes and candles and just for interest literally a couple of weeks ago there was a nice paper in frontiers treating the biosphere as a markov blanket so you know that there is practical and important sort of you know pressures in order to formalize what are the markov blankets of these large-scale structures and my guess is that the notion of wandering sets and the time scales over which internal states exchange with external states and internal states external states exchange with blanket states are going to be going to be an important issue and i say that because of course we develop what point when i i'm born and what point am i die does my can i what happens to my mark off blanket reproduction 3d printers printing themselves is a species of 3d printers one big markov blanket or do do actually drill down and just say markup only really exists the other markov blankets if we can imagine into the future maybe and if in the spirit of the free energy principle we maintain uncertainty about the free it may be the case that is we extend it to say wandering sets with wandering points we may develop a mathematics of more of a distance metric this cloud of points is currently this distance from a particular wandering point and they're orbiting around and there may be some dynamics in there and then if we follow a similar analysis we might end up back at the free energy principle plus another term for example like there may be an extra term over there which could be important would that be kind of a fair speculation of what might happen well yeah that's very clever yes yeah so i often say in moments of vanity and pride that the for perry and third world and in that should accommodate a judicious amount of uncertainty about itself is absolutely right yeah that's a very very clever observation in terms of the form i think that's probably right as well i mean you're asking me to think about you know where we might be in in say 10 years time and so this is just hand waving but certainly one could action applied not to probability distributions over states but probability distributions over over partitions uh and sets and getting into some things like category theory and the like so one can see a sketch or an image of the future kind of maths that there might be that might conserve the relatively simple principles that underwrite the free energy principle but now applied at some one level up as so not to the actual system itself but to the probabilistic configurations and the partitions that define that principle if i can zoom out a little bit here so the free energy principle is a lot and but there are situations in game theory and decision theory where that's fitness enhancing so a few examples here you might not want to learn the face of your kidnapper because he's more likely

to kill you and there's the other thing where you might so this is an information you don't want to gain and sometimes it's also useful to have false beliefs for example it can be useful to think you're smarter than you actually are or more confident than you actually are because that will help you in social situations yeah that's a fascinating area and i see those those sort of themes in many different contexts wishful thinking deliberately avoiding certain sources of information not wanting to open a letter that tells you whether you've passed your exams or a letter from the doctor that contains your the results of your recent scan for cancer for example there are things that we don't want to know so we're talking about very metacognitive things in a sense that i never heard that before about not a beautiful example it really does tell kind of external states that you are dealing with are composed of sentient creatures like you and that they also have beliefs so now you get into the game of putting lots of active inference agents together and multi-agent scenarios and how you infer the degree of sophistication with which you infer the intentions of others and the belief states of others and all all that gets into these very deep and enriched generative models that themselves now represent the belief states of other active inference uh your optimization machines as it were one other level i think that understand these sometimes paradoxical or counterintuitive expressions of optimality is just to acknowledge that the whole of the bayesian brain the whole basis of indeed free energy is a statement about prior beliefs and as such prior beliefs are going to be in the context of self-evidencing necessarily optimistic so if you remember before i was reading prior beliefs about the outcomes consequent on the behavior as preferences and the only reason i say preferences is because if those are the kind of states that i typically find myself in so if i'm a physicist though the attracting set of my non-equilibrium steady state then it looks as if i will always be working towards these prior states so they look literally mathematically attracting in the sense of being an attracting set but also psychologically attractive and valuable and therefore the preferred worked towards so baked into a bayesian reading of the information theoretic imperatives for choice is an optimism you're always moving towards or trying to solicit evidence that confirms that you are an eternal immortal well-loved warm creature so it has to be like it can't be any other way manifestations of that i always find appealing because they're just statements of the fact it couldn't be any other way i have to believe myself to be an expert in this or very proficient in that in order to realize those preferred fantasies through action to actually secure the evidence yes i am that kind of thing without those priors without those optimistic priors you would not get the kind of active inference that underwrites our existence to come back you know to some of the phrases that tim was using to introduce their free we

had this very interesting point from conor about there may be game theoretic pressures let's say to drive down accuracy in in some sense there could also be and i'm not sure if this strays outside the free energy principle or not but implementation factors so the lesson is back to maxwell's demons and all the very sophisticated kind information engine analysis that people have done from then until now has shown that the recording and computation of information is an entropic process it generates and so even though the free energy principle does have a term in there which is the entropy of my model per se i'm not sure if that captures the entropy associated and i mean the thermodynamic entropy if you will associated with actually maintaining and operating that model so it may be the case for example that you reach a point where increasing my intelligence a little bit more cost me much more to operate and maintain than it actually buys me an improved accuracy for example where do you see that trade-off does or doesn't fit into the free energy i think the trade-off fits very neatly in fact you could say that was you know in large part constituent of the free energy principle in the following sense and again you preempted everything that i might have said about this but i'll say it again so the complexity that we were talking about so if you recall an elbow edmond slower bound in machine posterior and your prime belief distributions now that is a complexity cost it's also the the cost that underwrites uh salon off and universal computation computational formulations of generalized or universal intelligence it is the thing that drives compression in predictive coding or engineering applications of it it is the thing that jurgen smith uber would emphasize if he were part of our conversation that sort of complexity minimization so doing it in a minimally complex and a minimally maximally compressed way that that's the heart of it that's important because by landauer's and the janinsky equality that also directly one to one dictates the number of jewels that you'll expend in belief updating when you move from your prior to the posterior so recall the complexity is just the relative entropy between the prior and the posterior it is the degree to sensory evidence and therefore is exactly the changing of the erasure of bits of that can be measured in gnats in natural units of using natural logarithms that via landauer's principle has an exact thermodynamic cost in joules so when informationally efficient complexity minimizing schemes compression schemes from my perspective anything that minimizes variational free energy or belief updating the minimizes variational free energy you are also exactly talking about the most efficient way of doing it thermodynamically so what that means is if you want to build the best kind of computer one simple way of scoring how good your computer is is to take two machines that have the same accuracy and the one that uses less electricity is the best one and

that will also by definition provide the least complex um account and thereby minimize its uh minimize its free energy so i think it's a really important practical observation that what we're aspiring to here are really cheap and cheerful machines that can that can do do their sort of their synthesis just to make it a bit more concrete what principle term with the complexity because i completely agree and get the point that's the crux that's the most important term i'm wondering if maybe there should be a yeah oh a temperature parameter yeah that's a great question same space and in the same currency and on the same footing as your log prior preferences your rewards your utility your bellman-esque like imperatives is that there is a seamless exchange in terms of gnats natural units between the decomposition of your expected free this extrinsic utilitarian pragmatic one of the benefits of removing the alpha or beta or temperature coefficient on the the complexity versus the the accuracy that for the expected free energy now becomes risk and ambiguity is that now you can talk about reward in terms of gnats you can talk about how rewarding information is because they're just some arbitrary carving up of a single expression they expected for energy that seems to me important because that is if you like the sort of in terms of a dimensional or unitary unit analysis that is a thing that truly does dissolve the expiration exploitation dilemma and puts the value of information alongside the value of a money or the value of a fruit drop if you're doing animal experiments so absolutely not you really want to totally avoid any temptation to start adding ad hoc hyper parameters like temperature to this beautiful construct which explains everything the other answer is absolutely yes that in order to in order to say a free energy functional it's not that's why very carefully said hamilton's principles stationary action is just keeping it flat at steady state it could be very high it could be very low and that starts to beg the question what kinds of steady states are non-equilibrium are we really interested in simulating reproducing describing and if you get behi under the hood in terms of the relative entropies and the mutual informations that shape these steady states and in particular the the distributions over this markovian partition into inside and outside the blanket states it turns out that one important attribute or description of the state depends upon the way in which the active states increase mutual information between the internal and the external and if you parameterize up with a particular parameter and we'll call it alpha gets in exactly as you say as a knob on the expected complexity versus the or ambiguity so what that means is in a more general formulation of the of the free energy principle or certainly its application to understanding active inference there would be this extra parameter that really tells you you're dealing with systems that are exquisitely structured in the sense that they are they occupy a very small number of states they

could occupy in a particular way that is active that they actively construct their mutual information between the inside and the outside at the moment that's not in the literature so there are ongoing debates amongst the younger people who love the maths of this about in by acknowledging there are different kinds of non-equilibrium steady states and in so doing you have to really think about the relative importance for this steady state of technically the risk and the ambiguity and the sensory entropy awesome maybe i should have gone with omega that way it's the last parameter we ever need but thank you appreciate it absolutely fascinating there's been a lot of emphasis on a single all-encompassing learning principle this grand unified theory of the brain or indeed any system now um i've been speaking to one of our friends on the show he's given us two questions this guy is dr hari valpola he's an applied theoretical neuroscientist and he's the ceo of curious ai which is a reinforcement learning based startup and he's a very accomplished scientist actually and and he says it does make sense to claim that there's a unified cortical algorithm and anatomically the cortex is very uniform and although there are adaptations to specific tasks they seem to be variations of a single unified template and in biology function follows form however he says that to claim that everything follows from free energy minimization is just overly simplistic in his view he says it's obvious that for the core text goal-orientated learning and perception is crucially important he says the cortex isn't just learning and perceiving all the structure and information it receives from the senses learning is heavily modulated by task demand and attention and so is perception itself so he gives this example he says phonemes of your mother tongue and you play some speech from the radio the result is nothing but if you play the same speech during a social interaction i think it's an excellent observation and it takes us into a sort of world which we haven't really been discussing so far which is the different flavors of free energy minimization and how it might be manifest and so just coming back to the point that structure matters structures of the brain being very finely attuned to and adapted to the context in which they're that they're making their inferences one would normally cast that in terms of structure learning if i was a statistician there are at least three if not four levels of maximizing model evidence i.e minimizing variational free energy also known as marginal likelihood i can make inferences to reduce my uncertainty about states that change with time i can do learning which is reducing or increasing the marginal likelihood of the parameters of my generative model i could also think about the hyper parameters that encode that usually cast in terms of precision or neg entropy of various distributions and that interestingly and speaks to ignoring things that we're talking about previously with conor sometimes it's best to ignore in a base optimal sense

if you've got very inconsistent and compatible information that's you know one base optimal explanation for ignoring and that speaks to sort of intermediate temporal scale of free energy minimization but the one i want to get to beyond that is structure learning also known as so if i was a statistician i wanted to score the quality of my generative model my hypothesis my convolution model whatever it was then i would score it with a free energy approximation or bound on the marginal likelihood of the model evidence now take another model another hypothesis and i would compare the adequacy of these two hypotheses using procedural bayesian model comparison and that is formally identical to choosing the hypothesis with the least free energy so the free scoring the evidence for this hypothesis of that hypothesis so you have now a mathematical description of what in cognitive signs would be known as structure learning optimizing the very structure of your model does it have three layers as my deep network fit for purpose should i add another layer how many units do i have in each layer do i have a more sparse connectivity do i use linear rectifying all of these structural aspects can in principle be evaluated in terms of the the free energy or the marginal of the underlying marginal likelihood or the evidence so you have this world of structure learning now that world itself can proceed at different time skills it can proceed in terms of the development of a particular architecture or phenotype if you're a developmental psychologist you would understand this structure learning from a neurodevelopmental point of view as the brain grows experience dependent that is driven or can be at least described by free energy minimization or self-evidencing or maximizing model evidence increa increasing through the addition of different connections different cortical layers and cortical areas or bringing them online through changes in connectivity that can actually progress right through until the early 20s in your development you would see that or you could describe that in terms of structure learning over somatic time but the exactly the same principles apply at an evolutionary time scale so this notion of free energy minimization as bayesian model selection speaks to a simple understanding of evolution or as nature's way of doing natural selection where you operationally associate the adaptive fitness with the probability this phenotype finding that phenotype in place and i should also say that all of these different expressions of free energy minimization at different spatial and temporal scales are all exactly consistent with some of the fundamental cybernetics and inception of theories of self-organization such as for example regulator theorem you know that it is on some reading provably true so what does that tell you it tells you that the structure including all those canonical micro circuits including those visual hierarchies of our brains for example or the multiple hierarchical layers in a deep neural

network have to be there if it is the case that the world that they're trying to has a deep structure if things are hierarchically generated and we know that the case for us because most of our world is caused by other creatures like us as we're currently witnessing so we know that there is that there is a hierarchical and dynamical aspect to it where the hierarchy say interpersonal interactions transcends just individuals but also groups of people con specifics families societies political affiliations theological affiliations or geographical hierarchical structure connectivity we find in a brain and in a variation auto encoder it really resonates with a lot of the things that we've been speaking about on on our podcast here because in machine learning we of course there are inductive priors built into models even the neural network has this kind of hierarchical organization which is an inductive prior but we've spoken a lot about meta learning as well when this thing that you're saying that there's a structurally relevant inductive pathway which could be learned but the last question from hari is he wants to know about the cerebellum for example he said there's plenty of evidence that it learns by supervised learning and can perfectly well handle regulation tasks without any models and he cites innate reflexes which train a complex feed-forward controller without any internal models i mean as a bit of a side thing here that valpolo is in a way implying that the free energy principle is is kind of unsupervised that's the way he's talking about which there might be some truth to that but we'll come back to that but he also cites the basal ganglia the hippocampus the amygdala and the superior caliculus each of them learns and their learning algorithms definitely don't seem to have much to do with the free energy minimization in his opinion and he says that the basal ganglia is a kind of reinforcement learning and the hippocampus is a one-shot learning and the amygdala and the superior caliculus are supervised learning for specific tasks and this is super interesting right because he cites kahneman system 1 and system 2 of thinking because their system two kicks in and they start getting a hundred percent accuracy very quickly but you don't see with monkeys so valpolo was saying that he doesn't see how that phenomenon could be explained by the free energy right again a whole raft of really interesting points there all of those things can be explained by the free energy principle and indeed there's a small literature on all of those things if people are interested you can get into the literature i have to say because the literature may not be the familiar literature for people in machine learning it's really he's asking some fundamental questions about functional anatomy and probably more specifically the computational anatomy that we can ascribe to various structures and hierarchies and connectomes in the brain so for example as the cerebellum watches the cortex and in particular the pre-motor cortex supplementary motor area and the

motor cortex per se hierarchically exchange messages all prescribing those predictions for that reflexive kind of predictive coding we're talking about before that are delivered to the spinal cord and the motor reflex arcs but of course that sort of low level description of motor control does rest upon this one deeply structured and informed hierarchy of predicting what am i going to do next because the cerebellum can watch that it can watch the cortex learning how to compose particular movements and if it can therefore learn how to do that then it can amortize it so i miss i'm assuming that we all know what amortization is um but what i what i mean here is that we can offset or defer the cost of the bayesian belief updating that we've talked about before is entails a computational complexity and therefore also a thermodynamic cost by actually hardwiring and learning the mappings from certain beliefs to certain sort of predictions say that drive murder reflexes in that sense the cerebellum can i think be very usefully understood in terms of being involved in supervised learning but i'd actually turn it on its head the cerebellum to tell the cortex well normally you do it like that why don't you just do it and efficiently and bake in those without having to worry about the the actual belief updating and converging to dynamically on a gradient flow to converging to a free energy minimum and that story i think can also be used to explain the particular role of say the dorsal and ventral striatum and the basal ganglia in arbitrating between whether we do a sort of system one versus a system do we think about stuff do we do our belief updating and a bit of planning as inference or do we just do what we've always done efficiently by harnessing something that has already interpretation of habitation the point of view of a neuroscientist would be the equivalent of system one versus versus system two in that sort of edge between machine learning and reinforcement learning sometimes referred to as model free versus model based and i would not subscribe to that in the sense that the the habit has to be learned from the the good old-fashioned belief updating and you can't you're not given habits you're given reflexes but to learn a habit to learn this is the kind of thing that i usually do in this situation and if i just do it i can dispense with the computational cost thereby minimizing the path integral free energy by doing the habit i noticed that slipped in there that because it's a principle of least action that entails a path integral so the amount of time it takes to minimize your free energy or get to your conclusion matters which means you have to do it quickly to minimize the action you've got to do it quickly which means there's a natural pressure to habitize and amortize in the service of minimizing free energy and i could go on in terms of telling you what i know about the functional anatomy of the amygdala but perhaps it's best just to say that there's a glorious game to be played here in fact it's the bread and butter of a

jobbing neuroscientist sonja systems neuroscientist to that has a deep structure so this is our job and it really started scratching the surface in the past decades with brain imaging and other tools at our disposal this is a very machine learning bias i'm not sure if you're familiar with this recently there has been a paper that says that predictive coding do you have any take on this yes it is contributed to that yeah yes no absolutely by coincidence i think we're having a an eurips workshop on saturday that sort of affords different perspectives on backprop i have to say from my perspective that's completely unsurprising and if there's any sense that prop has been demeaned as something which is not so from my it is invariably the case that the interesting things that basically is performing a gradient flow on a variational free energy functional then you also believe that they are via the chain rule minimizing their prediction error so it's for me backprop is just if you like a a nice linearized chain rule-esque way statement of things that work flow on free energy gradients that can always be written down with the back propagation of prediction errors it doesn't surprise me the slight is that predictive coding belongs to this class of algorithm as does propagation of errors and i would actually submit everything that works should be should be described in along those lines i've heard neuroscientists actually say something positive about backprop so that's really fascinating so i have a second question and that is in your own personal opinion what has been the most interesting progress in systems neuroscience you've seen since you first proposed the free energy principle whether it's directly related to it or not standing back from a non-personal point the whole pragmatic term towards embodiment we've seen in the 10th century in cognitive science has really been a very big thing that has been underwritten and also has been lent momentum by theoreticians like ourselves within that the role of predictive processing that's become a meme now so you can't write a paper in cognitive neuroscience now without a nod to predictive processing at some level of an inactive and embodied situated contextualized sword so that's been a slow and really important shift which is presenting all sorts of really issues in terms of exciting developments in the neurosciences an increasing quest for understanding the nature of these epistemic affordances which drive us in particularly in terms of social neuroscience echo niche construction communication with others and how we forage for information you know that seems to be in the context of usually social interactions also touching upon an understanding of feelings and emotion and how resolving uncertainty about bodily states in my world that will be interceptive inference gets into this game as an integral part of these deep hierarchical models that try to explain everything what role do literally gut feelings have in my propositional beliefs about where i am or who i am and how i should be

behaving so challenging questions that are now starting to address key aspects of intelligence from the point of view of neuroscience such as the emergence of selfhood and what underpins it and self in relation to others so we're now talking about cultural economic construction and multi-agent interactions so that's been very exciting so i guess maybe to end the show i'd like to ask you on potentially a controversial question i don't know i've been trying to find your official answer on this and i haven't been able to find it but if we take for granted that the free explains a great deal about how certain of the kind we've been elucidating will does it explain the origin of those so does the free energy principle for have anything to say about abiogenesis on earth on the origin of life on earth that first spark that created systems which can then follow a trajectory guided by the free energy principle or is that outside of its scope i'm sorry to add on a deflationary thought but i'm afraid it's outside of its scope i say that sort of in a rigorous way because remember all of the interesting math and all this physics of sentience all this bayesian mechanics all inherits from the assumption of a that entails a markov blanket as such where it's only about steady states it's not where that steady state came from so there's lots of there are lots of really interesting questions there's a great paper by kate jeffries and there's lots of work by people at los alamos around the world trying to say is it almost unavoidable delicate mock-off-blanketed systems great to know the answer to that and to have a tractable mathematical formulation of that this is the property these are the properties that it must possess but at least it can tell us we should continue exploring and looking around to address that uncertainty yes clever yes of course some better better free energy principle professor fristen thank you very much for joining us this evening it's been an wonderful talking to you thank you very much indeed thank you so much thank you truly a pleasure i'm going to be honest with you after talking to him myself i think he has a very intuitive understanding of very deep topics and a massive depth of knowledge and therefore when he tries to communicate i think he just what's the right way to say this he leaves behind a trail that's a little bit hard for us to follow because he's just moving at a very kind of deep and fast rate in my opinion he was trying to help us understand just operate on such a high that it's not always easy to follow tell them that that's just the result of externalized intelligence oh bad you might get your alpha constant i would have called it something like kappa after my first name alpha's good i mean start with the first one like why why do we need to go that's good it's the maximally ambiguous answer that still fits the data yeah yeah i when i at this hedge fund i worked out we had a problem one day and lost a couple million dollars or something not me personally but the fun and we we called up one of the guys who maintains our market

data feeds and we're like so was the ticker that had the problem today the nasdaq ticker or the nicey ticker and he goes yes we're like uh it's not a yes or no question so that was kind of an answer like from from person there it's like do we need alpha you know no and yes no anyway we really hope you've enjoyed the episode today we've had so much fun next week